Samuel Suvarnaraj Chettipalli

Controls & Automation Engineer

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SUMMARY

Controls and automation engineer (MSEE, University of Dayton) who programs ladder-logic sequencers for Allen-Bradley and Siemens PLCs, with defined fault handling from emergency stop to watchdog timeout to recovery, and builds the operator screens that show the machine state. Commissioning and troubleshooting experience from a controls internship at Loram Rail Maintenance. Seeking an entry-level controls or automation engineering role.

TECHNICAL SKILLS

PLC & Automation	Allen-Bradley Micro800, CCW, Siemens S7-1500, TIA Portal, S7-PLCSIM, Ladder Logic, Structured Text, PLC and I/O troubleshooting, interlocks
HMI & Simulation	Siemens WinCC, Factory I/O, Micro800 simulator, Ignition (learning)
Familiar with	Studio 5000, RSLogix 500, FANUC and KUKA robot basics, robot I/O, FactoryTalk View, MES
Electrical	Electrical schematics, wiring diagrams, 24VDC and 120VAC control circuits, motor controls, VFDs, PowerFlex, servo drives, relay logic, sensors, multimeter diagnostics, EPLAN Electric P8 (basic)
Conveyance & Mechanical	Conveyors, sortation systems, material handling equipment, mechanical drawings, electromechanical systems, equipment installation, repair and rebuilds
Maintenance & Safety	LOTO, 5-S, Arc Flash, NFPA 70E, preventive maintenance, root cause analysis, equipment reliability, downtime reduction
Networks & Tools	EtherNet/IP, Modbus TCP, Git and GitHub

PROJECTS

Pick and Place Sequencer — Siemens S7-1500, TIA Portal, S7-PLCSIM, Factory I/O

- Wrote a 10-step ladder state machine (26 networks, one memory bit per step) for a pick-and-place arm with no end-of-travel sensors, detecting each move as started, then stopped.
- Built Stop, E-stop (normally closed) and 20 s watchdog fault paths, each tested live in Factory I/O with Reset and restart recovery; found and fixed two sequencing bugs.
- Published the ladder printout, tag list, test results and demo videos in a public repository.

HMI Operator Screen for the Pick and Place Cell — Siemens WinCC, TP1200 Comfort

- Built a display-only WinCC screen showing both belts, the gripper position and state, and Running, Idle, Fault, Arm Lowered and E-Stop OK lamps, all driven by live PLC tags.
- Separated fault causes on screen: a watchdog stall turns the Fault lamp red while E-Stop OK stays green, while a pressed E-stop turns both red.
- Ran six tests against the live simulated cell (normal cycle, Stop, E-stop, Reset and restart, watchdog stall); recorded results and open items in a test plan.

Conveyor Sorting Station with Dynamic Buffering — Allen-Bradley Micro800, CCW, Modbus TCP

- Programmed a five-state conveyor sorting sequence in ladder (14 rungs, 21 variables) with photoeye detection and seal-in start/stop logic, plus congestion monitoring with three line-speed bands (30/60/100%) that divert to a buffer path on a bottleneck, reducing simulated jams by 35%.
- Built a latched E-stop safety path and TON-based jam and diverter-confirmation faults with manual reset, so the operator sees which condition stopped the line.
- Wrote a parallel Structured Text version for comparison, connected the controller to a simulated plant over Modbus TCP, and tested every speed band, divert path, fault and recovery case.

EXPERIENCE

Controls Engineer Intern — March 2022 – April 2023

Loram Rail Maintenance, Bengaluru, India

- Responded to equipment alarms alongside senior engineers and troubleshot PLC, sensor and actuator faults by checking I/O status and wiring, which cut average diagnosis time by 20%.
- Wired part of a control cabinet from electrical schematics under supervision; every point-to-point check passed with zero rework before the cabinet moved on.
- Assisted with installing and commissioning I/O, interlocks and E-Stop circuits and helped verify each against the schematics, which improved first-pass test success by 15%.
- Documented each fault, its root cause and the corrective action for the team and engineers, which improved repeat-fault tracking by 25%.
- Followed LOTO and electrical safety procedures on every job with zero safety incidents.

EDUCATION

Master of Science in Electrical Engineering, University of Dayton — December 2025

Bachelor of Science in Electrical and Electronics Engineering, Cambridge Institute of Technology — May 2023